

# Hypercalcemia Etiologies

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- Excess PTH production
  - Primary Hyperparathyroidism –
  - Tertiary Hyperparathyroidism
  - Ectopic PTH secretion
- Malignancy
  - PTHrP – Squamous cell carcinomas
  - Lytic metastases

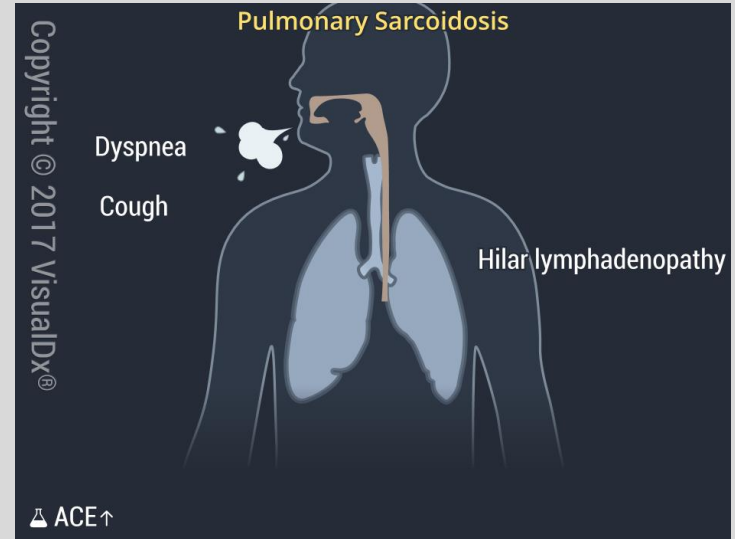


# Hypercalcemia etiologies

- Hypervitaminosis D
  - Sarcoidosis
- Excess Ca intake
  - Milk alkali syndrome
- Increase in bone reabsorption
  - Immobilization
  - Hyperthyroidism

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# Clinical Manifestations

- Correct for albumin
- Mild 11–11.5 mg/dL - asymptomatic
- Moderate 12-14 mg/dL – Fatigue, muscle weakness/pain
- Severe >14 mg/dL
  - Neuropsychiatric
  - Gastrointestinal
  - Renal & Cardiovascular

# Hypercalcemia Treatment

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- Treat underlying causes
- Hyperparathyroidism – parathyroidectomy if symptomatic. Asymptomatic based on risk of end-organ disease.

# Hypocalcemia Etiologies

- Low PTH (Hypoparathyroidism)
  - Destruction of gland
  - Abnormal gland development
  - Altered regulation of PTH

# Hypocalcemia Etiology

- High PTH (secondary hyperparathyroidism)
  - Vitamin D deficiency/resistance
  - Parathyroid hormone resistance syndromes
  - Extravascular deposition
  - Severe tissue injury

# Treatment

- Mild (7.5 to 8.0 mg/dL)/chronic hypocalcemia – oral calcium supplementation with vitamin D3
- Severe ( $\leq 7.5$  mg/dL)/symptomatic – IV calcium 90 to 180mg then switch to oral
- Correct concurrent hypomagnesemia

# Phosphate Regulation

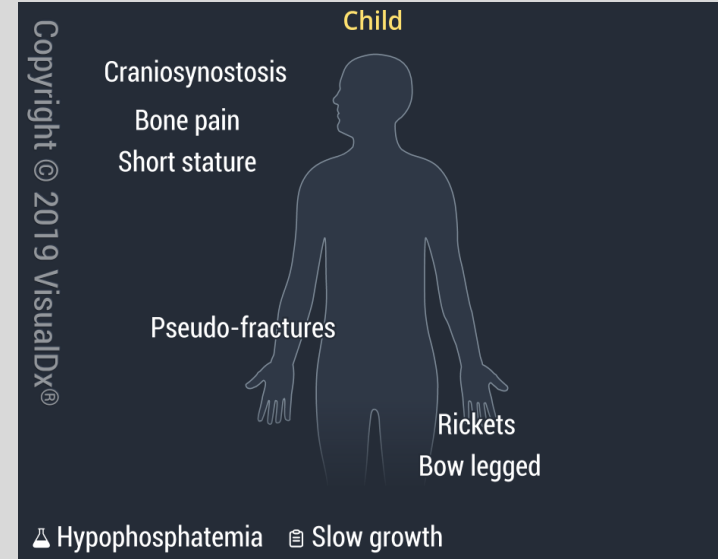
- Determined by rate of renal tubular absorption.
- Intestinal absorption highly efficient therefore regulated by urinary excretion.
- The fractional excretion of phosphate (ratio of phosphate to creatinine clearance) is generally in the range of 10–15%

# Phosphate Regulation

- Reabsorption depends on dietary intake, serum concentration, PTH, Vitamin D3 levels and FGF-23
- PTH and FGF-23 reduce phosphate absorption
- Low intake and serum levels increase absorption
- Main factor stimulating PTH is calcium not phosphate

## Reduced renal phosphate absorption

- Hyperparathyroidism
- Vitamin D deficiency
- Renal phosphate wasting – associated with rickets and osteomalacia
- Fanconi syndrome



## **Decreased intestinal absorption**

- Inadequate intake
- Antacids with aluminum or magnesium
- Steatorrhea or chronic diarrhea

# Diagnosis and Treatment

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## 24 hour urine phosphate excretion

- $<100\text{mg}$  – internal redistribution or decreased intestinal absorption
- $>100\text{mg}$  – reduced renal phosphate absorption

## Treatment –

- Asymptomatic  $<2.0\text{mg/dl}$  – oral therapy
- Symptomatic  $1.0\text{-}2.0\text{ mg/dl}$  - oral therapy
- Symptomatic  $<1.0\text{mg/dl}$  – IV therapy

## Impaired renal phosphate excretion

- Hypoparathyroidism
- Parathyroid suppression
- Renal insufficiency
- Acromegaly
- Familial tumoral calcinosis



## Question

An 88-year-old male presents to ED unconscious. While taking his blood pressure you notice carpal spasm. Which of the following is the most likely diagnosis?

- A. Hyperkalemia
- B. Hyponatremia
- C. Hypercalcemia
- D. Hypocalcemia
- E. Lambert–Eaton myasthenic syndrome

# Question

Lab studies are:

- Calcium 1.0 mg/dL (normal, 8.8-10.2 mg/dL);
- Phosphate 13.0 mg/dL (normal, 2.5-4.5 mg/dL)
- Parathyroid hormone 94.3 ng/L (normal 10-65 ng/L)
- 25-Hydroxyvitamin D 71 ng/mL (normal, 20-100 ng/mL)
- BUN 10 mg/dL (normal, 7-20 mg/dL)
- Creatinine 1.0 mg/dL (normal, 0.6-1.2 mg/dL)

**What is the most likely diagnosis?**

- A. Primary hyperparathyroidism
- B. Vitamin D deficiency
- C. Chronic kidney disease
- D. Squamous cell carcinoma of the lung
- E. PTH resistance syndrome

Labs are:

- Calcium 14.3 mg/dL (normal, 8.8-10.2 mg/dL)
- Phosphate 1.0 mg/dL (normal, 2.5-4.5 mg/dL)
- Parathyroid hormone 4.6 ng/L (normal, 10-65 ng/L)
- Parathyroid hormone-related peptide 3.8 pmol/L (normal < 2.0 p/mol)
- 1, 25-dihydroxyvitamin D 60 ng/mL (normal, 20-100 ng/mL)

**What is the most likely diagnosis?**

- A. Primary hyperparathyroidism
- B. Familial hypercalciuric hypercalcemia
- C. Multiple myeloma
- D. Squamous cell carcinoma
- E. Excess dietary calcium intake

# Magnesium Regulation

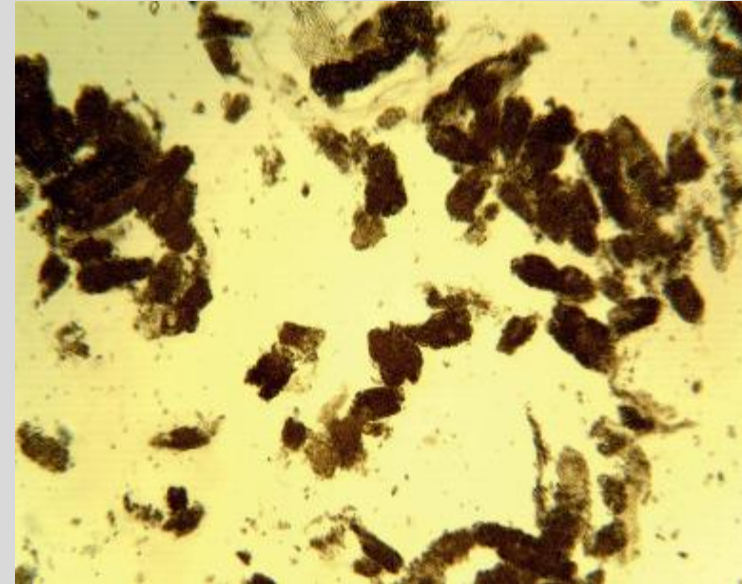
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- Thick ascending limb
  - Passive
  - Junctional proteins
  - Increased reabsorption with PTH and hypomagnesemia
  - Decreased reabsorption with hypercalcemia or hypermagnesemia
- Distal convoluted tubule
  - Important for fine tuning

## Renal losses

- Genetic wasting syndromes
- Medications
- Tubular dysfunctions
- Uncontrolled diabetes
- Hypercalcemia



# Diagnosis and Treatment

- If no apparent etiology check magnesium excretion.
- Mild asymptomatic – oral magnesium salts
- Severe – IV magnesium (i.e. torsades)
- Correct concurrent deficiencies

| Hypomagnesemia          |                                      |                   | Normal          | Hypermagnesemia   |                                          |                              |                                |
|-------------------------|--------------------------------------|-------------------|-----------------|-------------------|------------------------------------------|------------------------------|--------------------------------|
| Severe<br>←<br>< 0.5    | Moderate<br>0.5 - 0.9                | Mild<br>1.0 - 1.4 | 1.5 - 2.0 mEq/L | Mild<br>2.1 - 5.0 | Moderate<br>5.1 - 10.0                   | 10.0 - 15.0                  | Severe<br>→<br>> 15.0 mEq/L    |
| Cardiac arrest, seizure | ECG changes, tremor, spasm, cramping | Asymptomatic      |                 | Asymptomatic      | ECG changes, dizziness, nausea, weakness | Paralysis, areflexia, low BP | Respiratory and cardiac arrest |

# Question

Which of the following are expected manifestations of hypermagnesemia?

- A. Decreased deep tendon reflexes
- B. Refractory hypotension
- C. Respiratory failure
- D. Facial flushing
- E. All of the above